

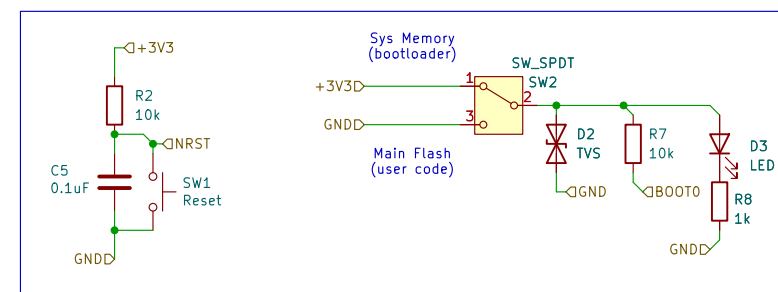
*Place oscillator circuitry close to MCU

Beware, using PC13 – 15 as GPIOs ...

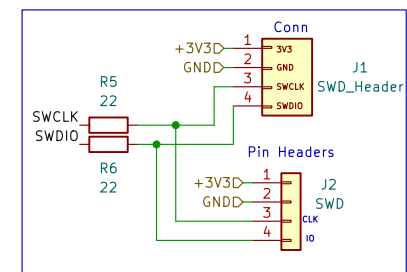
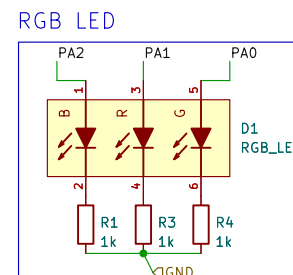
2. PC13, PC14 and PC15 are supplied through the power switch. Since the switch only sinks a limited amount of current (3 mA), the use of GPIOs PC13 to PC15 in output mode is limited:
 - The speed should not exceed 2 MHz with a maximum load of 30 pF
 - These GPIOs must not be used as current sources (e.g. to drive an LED).
3. After a backup domain power-up, PC13, PC14 and PC15 operate as GPIOs. Their function then depends on the content of the RTC registers which are not reset by the system reset. For details on how to manage these GPIOs, refer to the Backup domain and RTC register descriptions in the reference manual RM0440 "STM32G4 Series advanced Arm®-based 32-bit MCU".



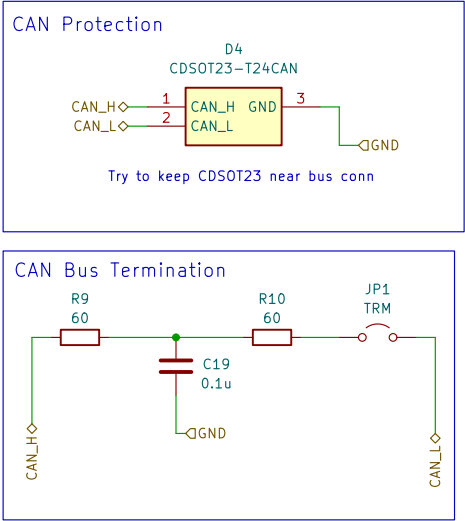
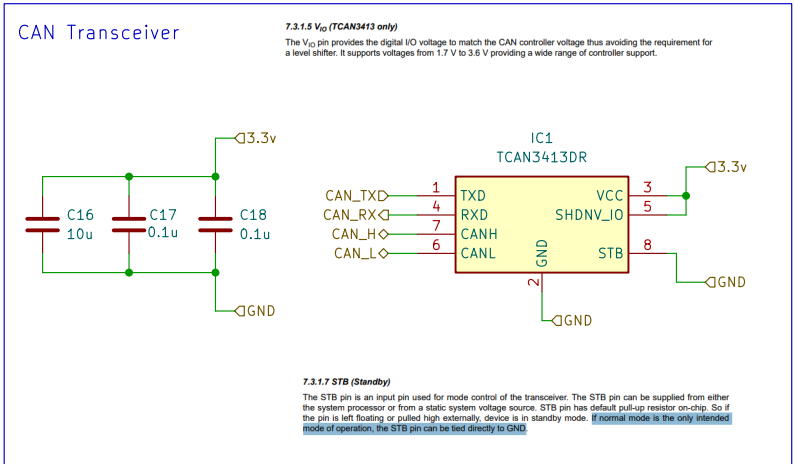
Reset BTN + Boot SW

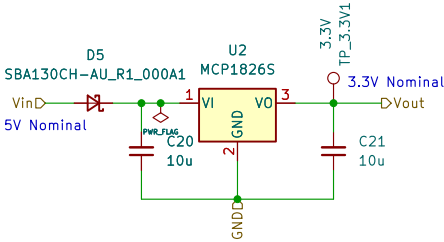


Serial Wire Communication

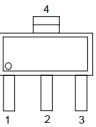


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SOT-223-3



Pin	
1	V _{IN}
2	GND (TAB)
3	V _{OUT}
4	GND (TAB)

Parameters	Min	Nominal	Max	Unit
Input Operating Voltage	2.3		6.0	V
Input Quiescent Current		120	220	uA
Max Output Current	1000			mA
Line Regulation		+−0.05	+−0.20	%/V
Load Regulation	−1.0	+−0.5	1.0	%
Voltage Regulation	VR−2.5%	VR+−0.5%	VR+2.5%	V
Output Short Circuit Current		2.2		A
Dropout Voltage		250	400	mV

VR=regulated output voltage=3.3V

Input Nominal: 5V
Diode Forward Voltage: 500–600 mV
LDO Dropout Voltage Max: 400 mV
−> −0.7V headroom on input

Sheet: /LDO 3.3V/
File: LDO.kicad_sch

Title:

Size: A4 Date:

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$$C_{dvdt} = \frac{I_{GATE} \times C_{OUT}}{I_{INRUSH}}$$

$$C_{dvdt} = \frac{(60 \cdot 10^{-6}) \cdot (200 \cdot 10^{-6})}{1} = 1.2 \times 10^{-8}$$

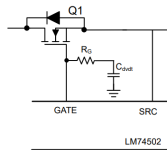


Figure 8-2. Inrush Current Limiting Using LM74502

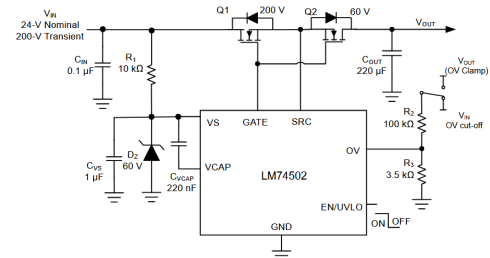
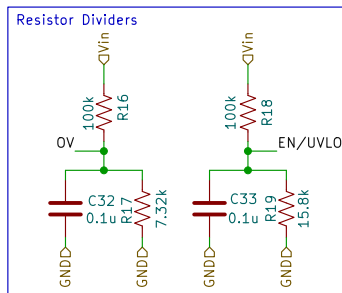


Figure 9-8. Typical Surge Stopper Application for 24-V Powered Systems

$$V_{uv} = V_{um} \left(1 + \frac{R_{bot}}{R_{top}} \right) \quad V_{OVR} = \frac{R_2 \times V_{OV}}{R_1 + R_2}$$

Output of Python Calculations for Resistor Values

	Min	Typ	Max
Vov Falling	15.29	16.76	18.25
Vov Rising	16.77	18.33	19.91
Vuv Falling	7.40	8.36	9.21
Vuv Rising	8.36	9.09	9.84



Kell Harpole

Sheet: /PowerProtection/
File: PowerProtection.kicad_sch

Title: Power Protection

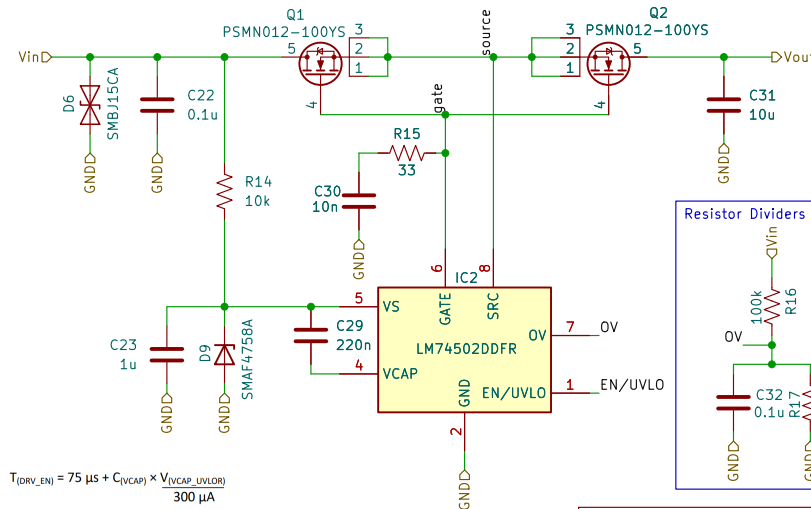
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Rev:

Id: 5/10



$$T_{(DRV_EN)} = 75 \mu s + C_{(VCAP)} \times \frac{V_{(VCAP_UVLOR)}}{300 \mu A}$$

4.84ms = 75us + 220nF * 6.5V/300uA
6.5V recommended on datasheet
4.84ms power up delay

$$P_{DZ} = V_{DZ} \times \frac{(V_{IN(MAX)} - V_{DZ})}{R_1}$$

$$P_{DZ} = 58 \cdot \frac{(120 - 58)}{10000}$$

$$= 0.3596$$

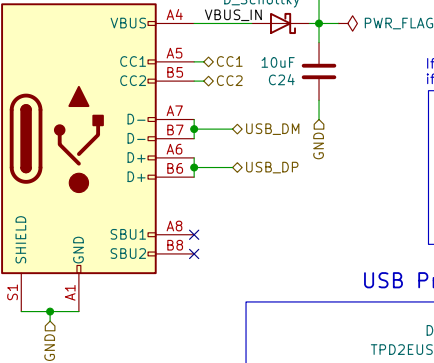
$$P_{R1} = \frac{(V_{IN(MAX)} - V_{DZ})^2}{R_1}$$

$$P_{R1} = \frac{(120 - 58)^2}{10000}$$

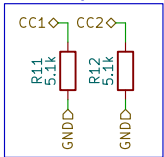
$$= 0.3844$$

USB-C Port

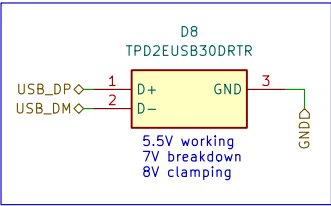
J3
USB_C_Receptacle_USB2.0_16P



If using USB-C PD, remove CC1 and CC2 pull down resistor,
if not using USB-C PD, keep them



USB Protection



Sheet: /USB-C_Connector/
File: USB-C_Connector.kicad_sch

Title:

Size: A4

Date:

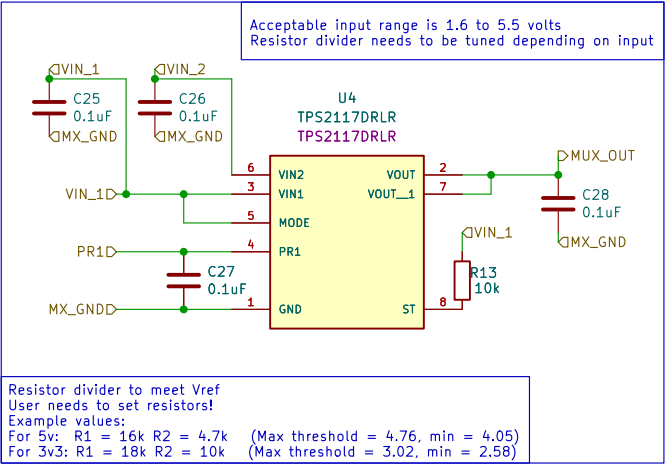
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Rev:

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5V/3V3 Power Mux

Configured to use automatic switching mode: VIN_1 is prioritized



Max and min threshold voltage calculations for 3v3 and 5v
[delete this if you want :)]

3v3

$R_1 = 18000$

$R_2 = 10000$

$M = 1.08 \frac{(R_2 + R_1)}{R_2}$

$m = 0.92 \frac{(R_2 + R_1)}{R_2}$

= 3.024

= 2.576

5v

$R_1 = 16000$

$R_2 = 4700$

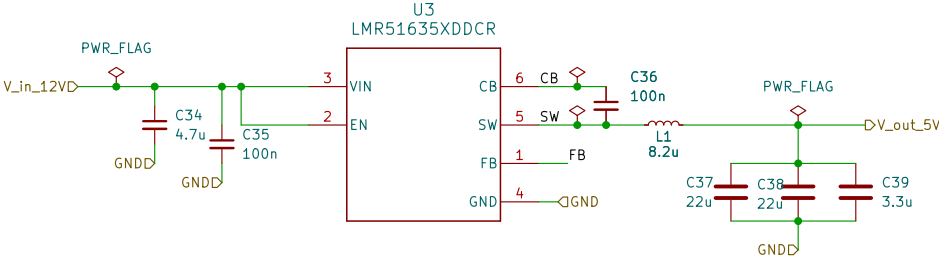
$M = 1.08 \frac{(R_2 + R_1)}{R_2}$

$m = 0.92 \frac{(R_2 + R_1)}{R_2}$

= 4.75659574468

= 4.05191489362

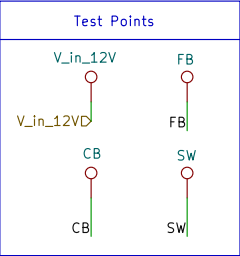
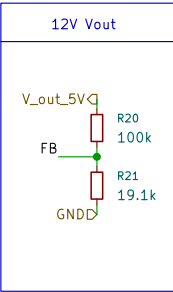
12V-5V Buck Converter



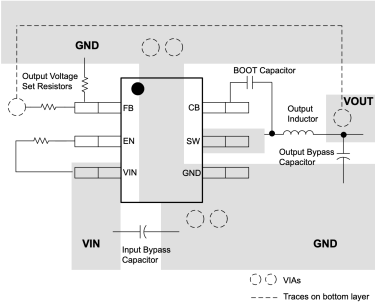
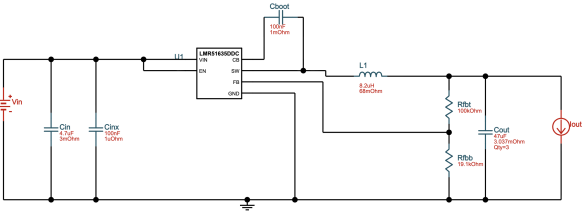
Fsw 400khz Vout 5

Table 8-2. Typical External Component Values for LMR51635, 3.5A Output Current

f _{sw} (kHz)	V _{OUT} (V)	L (μH)	C _{OUT} (1)	R _{FBT} (kΩ)	R _{FBB} (kΩ)	C _{FF}
400	3.3	5.6	2 × 47μF / 10V	31.6	10.2	Open
	5	10	2 × 33μF / 16V	53.6	10.2	Open
	12	15	10μF + 22μF / 35V	143	10.2	Open
1100	3.3	2.2	2 × 22μF / 10V	31.6	10.2	Open
	5	3.3	33μF / 16V	53.6	10.2	Open



8.4.2 Layout Example



Sheet: /12VBuck/
File: 12VBuck.kicad_sch

Title:

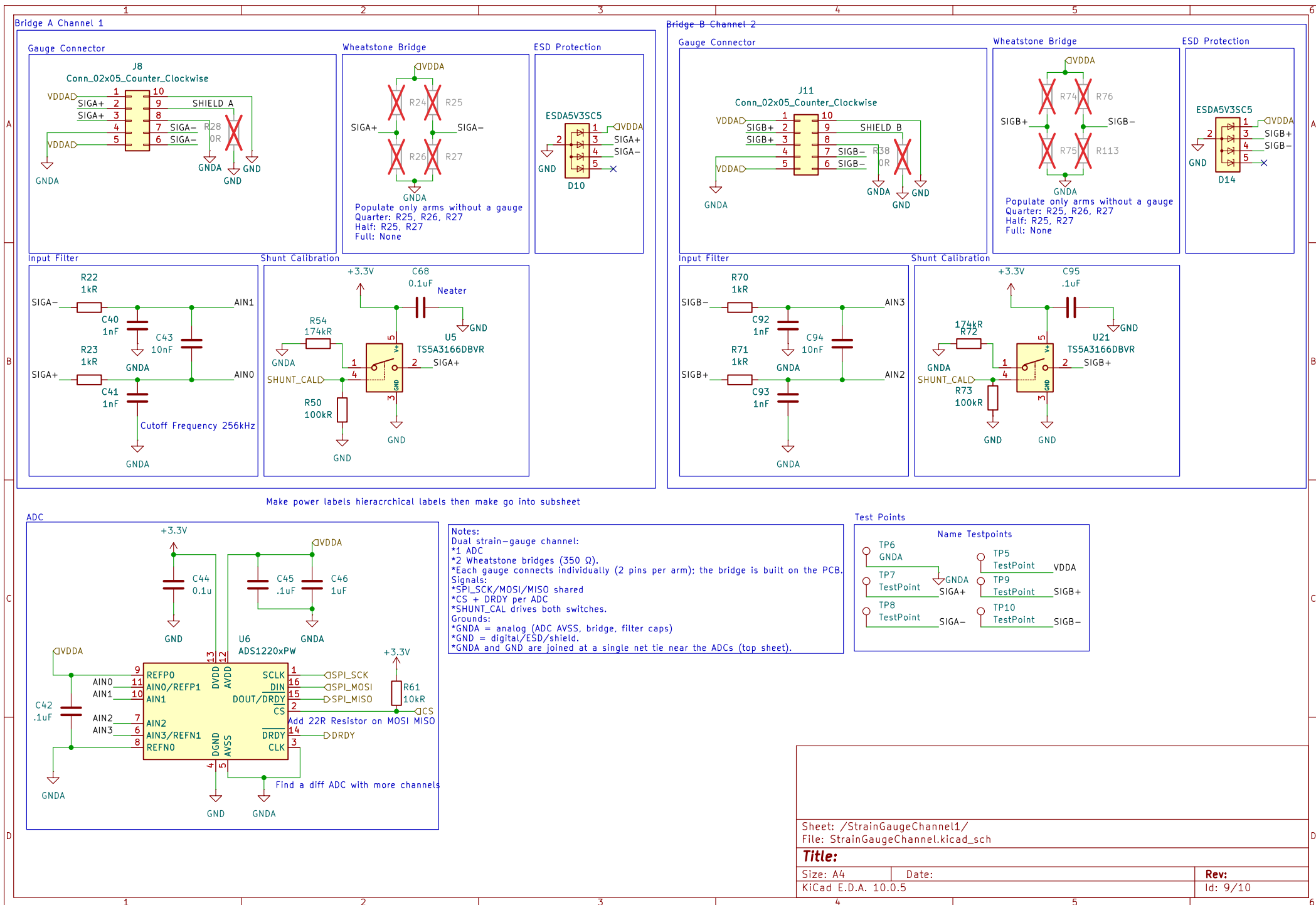
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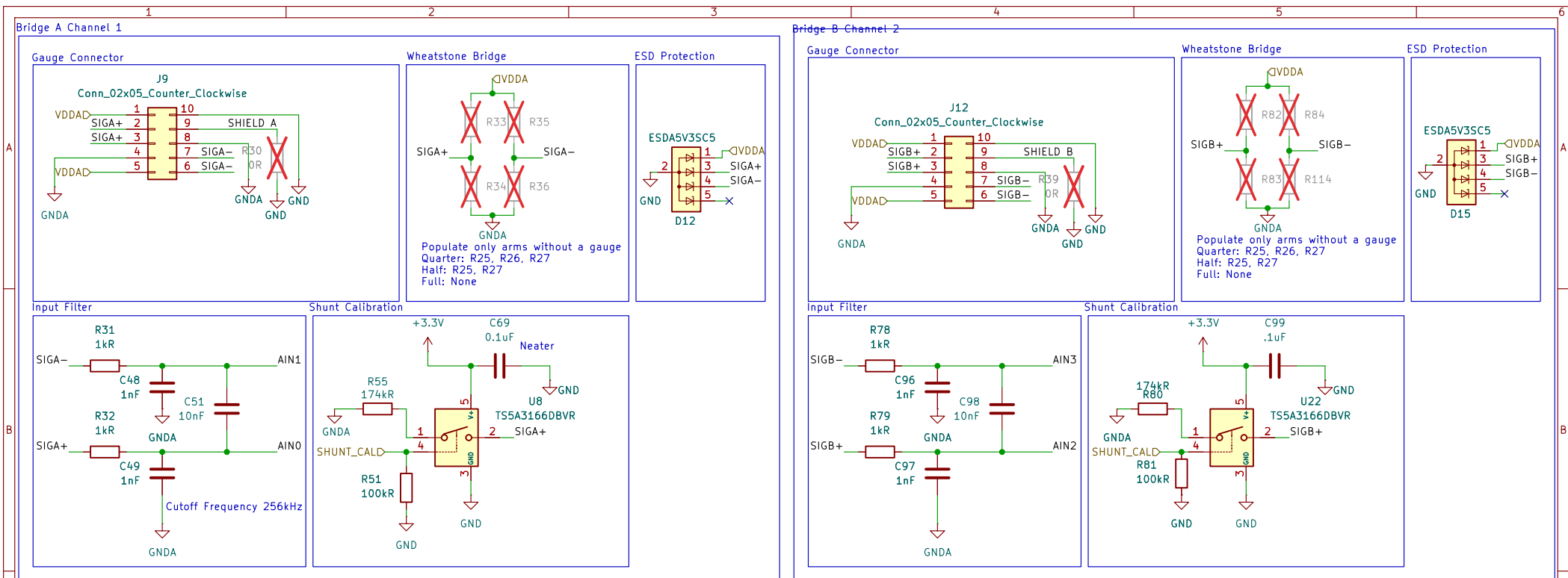
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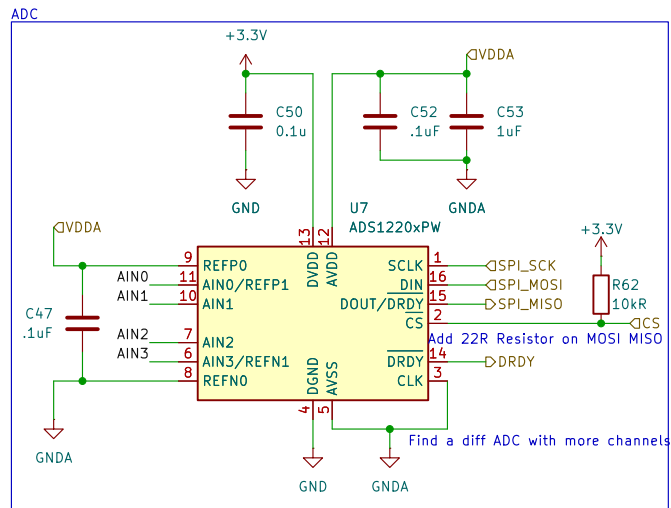
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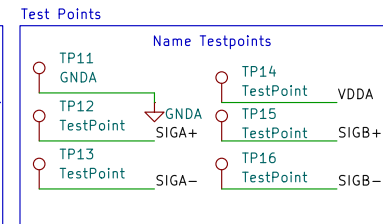




Make power labels hierarchcal labels then make go into subsheet



Notes:
Dual strain-gauge channel:
*1 ADC
*2 Wheatstone bridges (350 Ω).
*Each gauge connects individually (2 pins per arm); the bridge is built on the PCB.
Signals:
*SPI_SCK/MOSI/MISO shared
*CS + DRDY per ADC
*SHUNT_CAL drives both switches.
Grounds:
*GND_A = analog (ADC AVSS, bridge, filter caps)
*GND_D = digital/ESD/shield.
*GND_A and GND_D are joined at a single net tie near the ADCs (top sheet).



Sheet: /StrainGaugeChannel2/
File: StrainGaugeChannel.kicad_sch

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